

AGROVA

Your AI Farm Companion

Hackathon Project Concept Document

1. Executive Summary

Agrova is an AI-powered farm intelligence platform designed to help African farmers manage livestock and make better farming decisions.

Farmers can interact with Agrova using natural language, voice and images. The system converts everyday farm activities into structured records, financial insights, livestock health assessments, reminders and early-warning alerts.

Agrova is designed to support poultry, goats, sheep, cattle, pigs, rabbits and other livestock as the platform expands.

2. The Problem

Many farmers still depend on notebooks, memory, spreadsheets and scattered WhatsApp conversations. This makes it difficult to consistently track feed, medication, vaccination, mortality, sales and profitability.

Digital adoption can also be affected by internet connectivity, data costs, digital literacy and language. Farmers need useful digital intelligence without having to become software experts.

3. Existing Solutions & Market Gap

Our investigation found that multi-livestock farm-management and AI-assisted agricultural platforms already exist, including KacePro, Farmwise, Cowtribe, FARMERS ERP and AgricSoft.

This means Agrova should not compete simply by being another dashboard for recording farms. The opportunity is the intelligence and accessibility layer: natural-language interaction, simple data capture, proactive anomaly detection and an AI layer connecting health, feeding and financial information.

4. Our Unique Angle

Agrova's central philosophy is: **"A farmer shouldn't have to learn software to use AI. AI should learn how the farmer farms."**

The core loop is: **TALK → RECORD → UNDERSTAND → ACT.**

Farmers communicate naturally; Agrova extracts structured farm information; AI analyzes the farm; and the system gives actionable recommendations and alerts.

5. Core MVP Features

Feature	Purpose
AI Farm Assistant	Natural text/voice interaction and automatic record extraction.
Smart Livestock Records	Animals/batches, purchases, feed, health, vaccination, mortality and sales.
AI Health Assistant	Symptoms and optional images → health-risk assessment and next-step guidance.
Farm Profit Intelligence	Expenses + sales + feed → revenue, costs and estimated profitability.
Vaccination & Task Reminders	Track upcoming farm activities.
Farm Early Warning	Detect unusual mortality, feed, production and expense patterns.

6. AI Farm Assistant

Farmers can type or speak naturally. Example: *“I bought two bags of feed today for ₦80,000.”* Agrova interprets the message and creates the appropriate record.

7. Multi-Livestock Management

Agrova can support poultry, goats, sheep, cattle, pigs, rabbits and other livestock. Poultry can be managed primarily by batches, while animals such as goats or cattle can also be tracked individually.

8. AI Livestock Health Assistant

A farmer can describe symptoms and optionally provide an image. Agrova can ask relevant follow-up questions and provide a possible-condition/risk assessment, recommended next steps and veterinary escalation guidance.

The feature is positioned as **AI-assisted livestock health assessment and decision support**, not a replacement for a veterinarian.

9. Feeding Management

Agrova tracks feed purchases, consumption, feeding schedules and feed costs. It can identify unusual patterns, such as unexpectedly high feed consumption.

10. Farm Financial Intelligence

Agrova tracks income from animal and farm-product sales and expenses such as feed, medication, vaccination, labour, transportation and equipment.

It can calculate revenue, expenses and estimated profit/loss and answer questions such as: *“How much have I spent on my goats this month?”* or *“Which livestock is making me the most money?”*

11. Vaccination & Task Management

Agrova can track vaccinations, deworming, medication and other scheduled farm activities and provide reminders.

12. Farm Early Warning System

This is one of Agrova's major intelligence features. Instead of waiting for the farmer to ask, Agrova monitors farm information for patterns such as rising mortality, unusual feed consumption, declining production, increasing medication use and rapidly rising expenses.

Example: **“Potential farm anomaly detected. Mortality in Broiler Batch B01 has increased significantly this week.”**

13. African-First Approach

The roadmap includes support for Nigerian Pidgin, Yoruba, Hausa, Igbo, Swahili and other African languages.

Agrova should also move toward offline-first functionality so farmers can record important information when connectivity is poor and synchronize later.

14. Target Users

Smallholder livestock farmers; commercial farms; mixed-livestock farmers; farm managers; agricultural cooperatives; and, in future, veterinarians, extension officers, feed suppliers, financial institutions and insurers.

15. Proposed MVP

1. AI Farm Assistant — text/voice interaction.
2. Smart Farm Records — natural conversation converted into structured records.
3. AI Health Assessment — symptoms and optional images.
4. Profit Intelligence — expenses, sales, feed and profitability.
5. Early-Warning Alerts — abnormal farm patterns.

16. Suggested Technology Stack

Frontend: React / Next.js

Backend: Python + FastAPI

Database: PostgreSQL

AI: LLM API for natural-language understanding, record extraction, farm analysis and recommendations

Vision: Vision-capable AI model for optional livestock image assessment

Voice: Speech-to-text and text-to-speech

Authentication: Simple phone/email authentication for the MVP

17. Basic System Architecture

Farmer → Agrova Interface → AI Engine → Backend API → PostgreSQL Farm Database → Analytics/Alert Engine → Actionable Insight → Farmer

The AI should not be treated as a standalone chatbot. It should work with structured farm data so its responses are grounded in the farmer's actual records.

18. Proposed Database

Users; Farms; Livestock/Batches; Health Records; Feed Records; Expenses; Sales; Vaccinations; Tasks/Reminders; Alerts; and AI Conversation/Action Logs.

19. Hackathon Demo Story

Step 1: Create a farm containing multiple livestock types.

Step 2: Speak a purchase/feed transaction naturally and show Agrova converting it into structured records.

Step 3: Ask Agrova for current spending and estimated profitability.

Step 4: Describe a livestock health problem and optionally upload an image.

Step 5: Show the dashboard updating with the new information.

Step 6: Demonstrate a farm anomaly alert based on mortality, feed or production data.

Step 7: Finish by showing that the farmer can manage the farm through conversation rather than complicated forms.

20. Why Agrova Can Stand Out

Natural conversation instead of form-heavy data entry; multi-livestock support; AI-assisted health intelligence; financial intelligence; proactive alerts; and an African-first approach around language, connectivity and digital literacy.

21. What We Should NOT Build for the MVP

Do not attempt a complete veterinary diagnostic system, huge marketplace, full accounting suite, every possible livestock species, every African language, complex IoT hardware or a large social network.

The MVP should prove one powerful concept: **Conversation** → **Farm Data** → **Intelligence** → **Action**.

22. Future Expansion

Agricultural marketplace; veterinary ecosystem; agricultural finance; insurance; IoT integrations such as smart scales and environmental sensors; and expansion from Nigeria to other African markets.

23. Business Model

Potential models include a freemium farmer plan, premium subscriptions, commercial farm plans, cooperative plans and B2B partnerships with feed companies, veterinary organizations, financial institutions and insurers.

24. One-Sentence Pitch

“Agrova is an AI-powered farm companion that transforms everyday conversations and farm activities into intelligent livestock management, financial insights, health assessments and early-warning alerts for African farmers.”

25. Team Roadmap

1. Confirm Agrova and the project direction.
2. Map the concept against the hackathon judging criteria.
3. Finalize MVP features.
4. Design the user journey and UI/UX.
5. Design the database schema.
6. Build frontend, backend and AI integration.
7. Test the conversation → record → insight → alert flow.
8. Prepare demo data, pitch deck and final presentation.

26. Long-Term Vision

Agrova should become more than an app: an intelligent digital operating system for African livestock farming.

The long-term vision is to give farmers a digital companion that understands their communication, remembers farm history, recognizes important changes, helps them make better decisions and makes modern farm intelligence accessible without requiring advanced digital skills.

AGROVA — Your AI Farm Companion

Built for farmers. Powered by AI. Designed for Africa.

The goal isn't to make farmers learn technology. The goal is to make technology understand farmers.